



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Skill Progress Analytics with Personalized Learning Path Recommendation Engine

Course Name :Java Programming

Course Code :CSA0925

Guided by

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By

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Review-1 Feedback

Feedback Received

- Need clearer system architecture and module flow.
- Need detailed explanation of database and UML design.
- Need to show implementation progress.
- Need to include UI and testing details.

Corrective Actions

- Updated the system architecture.
- Defined all major project modules.
- Added UML and ER diagram designs.
- Started implementation of Java and MySQL modules.
- Added UI, testing and reporting components.

Updated Architecture

Changes Made

- Added separate modules for student registration and login.
- Added course management and progress tracking.
- Added skill analytics module.
- Added personalized recommendation engine.
- Added dashboard and report generation.

Final Module Flow

- **Student → Login → Course Tracking → Progress Analysis → Skill Analytics → Recommendation Engine → Dashboard & Reports**

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MODULE DESIGN

1. User & Course Management Module

- **Input:** Student details, login credentials, course information
- **Process:** Validates users and manages student and course records.
- **Output:** Student profile and available course information.
- **Functions:**
 - Student Registration
 - Login Authentication
 - Course Management
 - Learning Activity Management
 - Data Storage

Module 2: Progress & Skill Analytics

1. **Input:** Quiz scores, completed courses, learning activities
2. **Process:** Analyzes student performance and calculates learning progress.
3. **Output:** Progress status, strengths, and weak areas.
4. **Functions:**
 - Quiz Score Recording
 - Course Completion Tracking
 - Progress Calculation
 - Skill Analysis
 - Strength and Weakness Identification

3. Recommendation & Dashboard Module

1. **Input:** Student progress and skill analysis results
2. **Process:** Generates suitable learning recommendations based on the student's performance.
3. **Output:** Personalized learning path, recommendations, and progress reports.
4. **Functions:**
 - Personalized Learning Path Generation
 - Course and Topic Recommendation
 - Weak Area Improvement Suggestions
 - Progress Dashboard
 - Report Generation



Module Output

Student Management

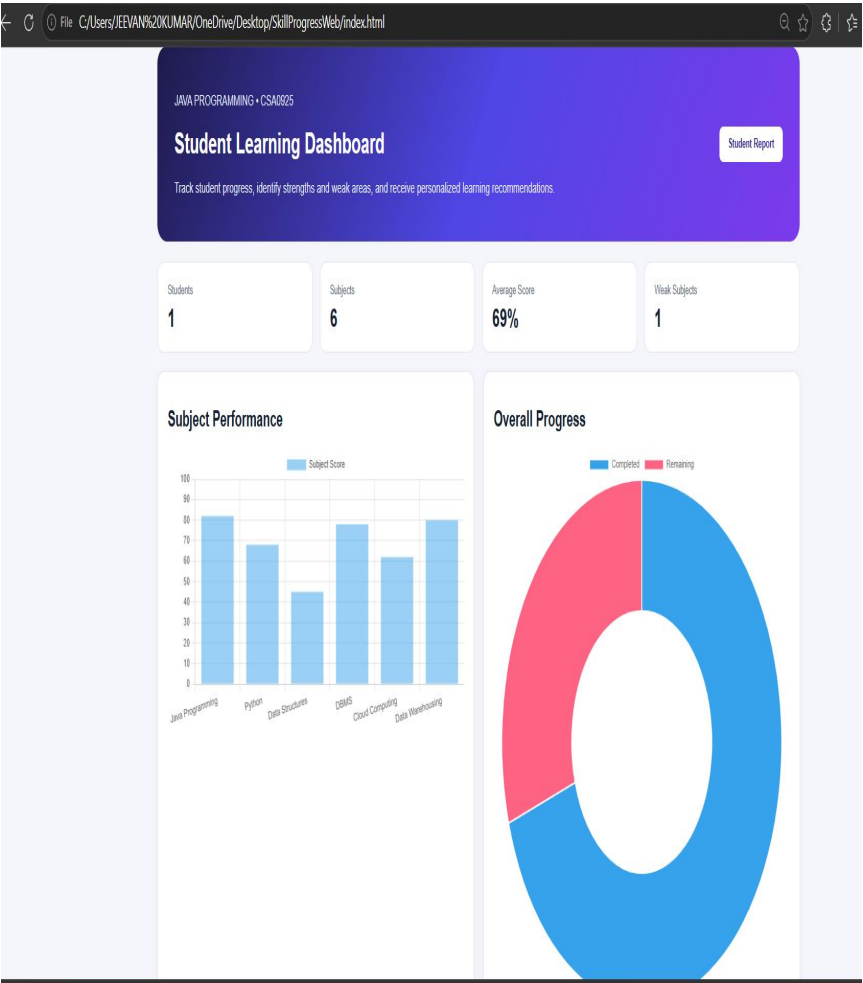
Student registration and student information

Add Student

Add Student

Registered Students

ID	NAME	EMAIL	DEPARTMENT	REPORT
1	P. Jeevan Kumar	jeevan@example.com	CSE	<button>View Report</button>



Student Report

Skill progress and personalized learning report

[Print Student Report](#)

Skill Progress Analytics

Personalized Learning Path Recommendation Engine

Course Performance

SUBJECT	SCORE	PROGRESS	STATUS
Java Programming	82%	85%	Strong
Python	68%	70%	Average
Data Structures	45%	50%	Weak
DBMS	78%	75%	Strong
Cloud Computing	62%	60%	Average
Data Warehousing	80%	80%	Strong

Skill Analysis

SUBJECT	SCORE	PERFORMANCE
Java Programming	82%	Strong
Python	68%	Average
Data Structures	45%	Weak
DBMS	78%	Strong



Engineer to Excel

UML & Database



Use Case Diagram

- Student Login
- Student Registration
- View Courses
- Track Progress
- View Recommendations
- Generate Reports

Class Diagram

- Student
- Course
- Progress
- SkillAnalytics
- Recommendation
- Report

Sequence Diagram

Login → Validate User → Retrieve Progress → Analyze Skills → Generate Recommendation → Display Result

ER Diagram

- Student
- Course
- Progress
- Recommendation
- Report

Technology Stack

Programming Language

Java

Database

MySQL

Connectivity

JDBC

Development Tools

NetBeans / Eclipse

Operating System

Windows 10/11

Hardware

Intel i5/i7 Processor

8 GB RAM

256 GB SSD

Implementation

Implementation Progress

- Student registration and login module implemented.
- Course management module developed.
- Progress tracking functionality developed.
- Database structure created using MySQL.
- JDBC connectivity established.
- Recommendation and reporting modules integrated.

Code Summary

- Java classes handle application logic.
- JDBC handles database communication.
- MySQL stores student and course information.

Completion:

Approximately 80–90%

If your actual completion percentage is different, use your real percentage.

User Interface

Login

- Secure student login.
- Registration for new users.
- User validation.

Dashboard

- Displays student progress.
- Shows skill strengths and weak areas.
- Displays recommended learning paths.

Reports

- Course completion report.
- Skill analysis report.
- Personalized recommendation report.

UI Flow:

Login → Dashboard → Progress → Recommendations → Reports

Testing

Unit Testing

- Tested login functionality.
- Tested registration functionality.
- Tested progress calculation.
- Tested recommendation module.

Integration Testing

- Tested Java–JDBC connection.
- Tested JDBC–MySQL communication.
- Tested data flow between modules.

Expected Outputs

- Correct student login.
- Accurate progress tracking.
- Identification of weak skills.
- Personalized learning recommendations.
- Progress reports.

Challenges & Future Plan

Challenges

- Maintaining accurate student learning data.
- Database connectivity issues.
- Integrating multiple Java modules.
- Time limitations during development.

Solutions

- Added input validation.
- Tested database queries and JDBC connections.
- Integrated modules step-by-step.
- Prioritized core functionalities.

Next Plan

- Complete final UI improvements.
- Perform complete integration testing.
- Fix remaining errors.
- Improve recommendation accuracy.
- Complete documentation and final deployment.

THANK YOU